Technical Report: Design and Structural Analysis of a 4-DOF Robotic Arm

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Abstract

This document describes the development of a 4-degree-of-freedom (DOF) robotic arm based on the Freenove base kit, which is an Arduino-like set for mechatronic applications. The project integrates the evolution from an initial physical prototype to a digital model validated through Finite Element Analysis (FEA), with a particular focus on interpreting the stress singularities that emerged during the simulation.

1 Project Objective and Hardware Components

The primary objective is the development of a fully functional 4-DOF robotic manipulator. The following components were used for actuation and kinematic control:

- **Joint actuators:** 3x SG90 9g micro servos (in the soon to be released v2 version they will be replaced by MG996R due to the very low SG90 critical load).



Figure 1: SG90 9G Micro Servo Motor

- **Base actuator:** 1x 28BYJ-48 step motor.



Figure 2: Unipolar stepper motor kit 28BYJ-48 and driver

- **Input interfaces:** 3x potentiometers.

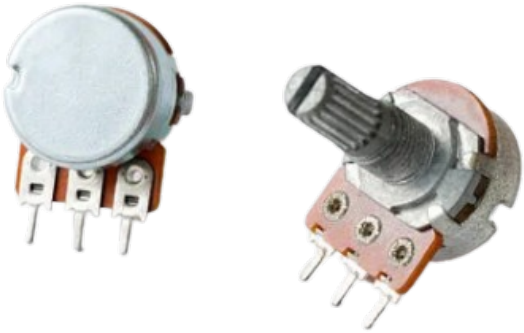


Figure 3: Potentiometers used for joint control

- **Input interfaces:** 1x joystick.

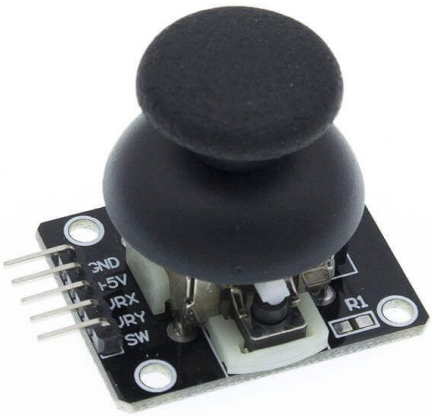


Figure 4: Joystick module

2 Development and Prototyping

The project started with the already released 2-DOF version made completely out of discard material such as toothpicks, cardboard, and pens.

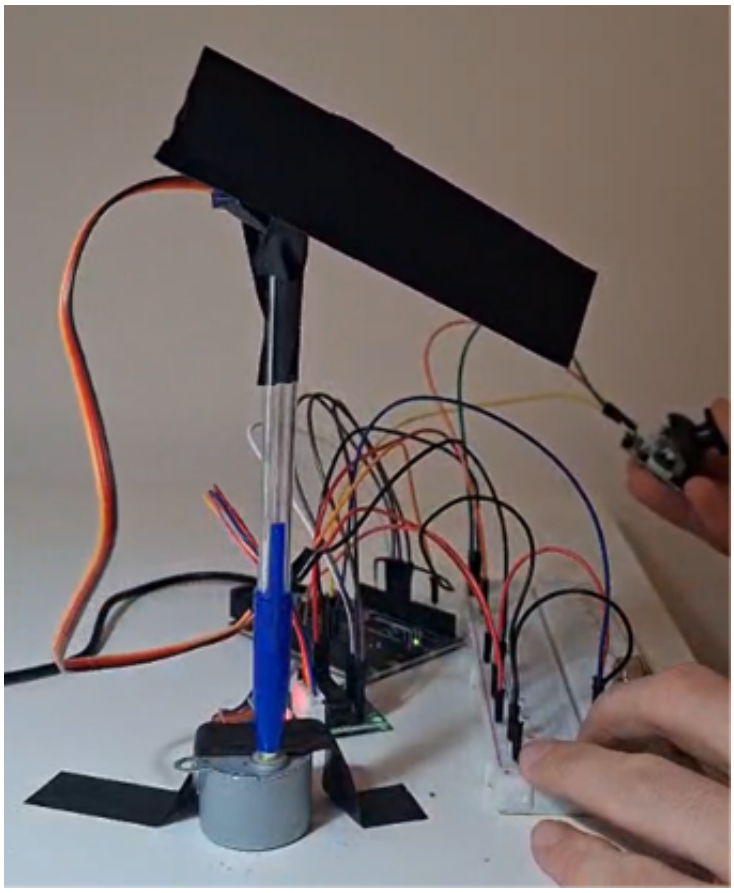


Figure 5: Initial 2-DOF prototype

The 2 added degrees of freedom came by adding 1 more movable arm and a gripping claw able to lift small loads. The first real model was also built using discard materials and gave a solid base for the 3D model I created; it featured a controlling board for each part of the arm (except for the claw) using a joystick and 2 potentiometers:

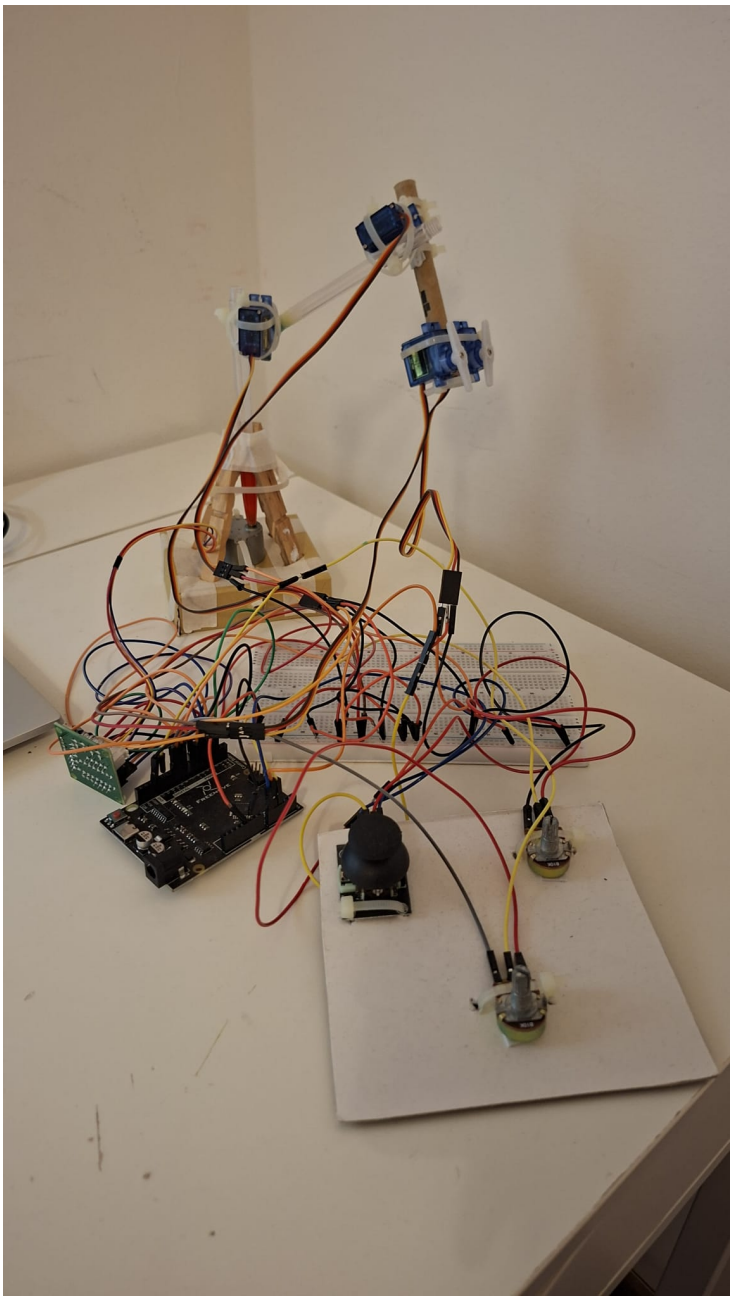


Figure 6: First physical model of the 4-DOF arm with control board

3 3D Modeling and FEA

I started 3D modeling each part of the arm starting from the base and going upward using Inventor and SolidWorks. Each piece was tested using FEA (Finite Element Analysis) through Inventor in order to make sure that it will be able to withstand the maximum load in the weakest possible position.

All the single part testing considered a load of 1 kg gripped by the claw, which is way more than the SG90 servo motors can really handle. Therefore, the mechanical part will break down before the plastic reaches the yield strength, preventing any irreversible plastic deformation.

Each single piece of the arm has its own folder in which you can find the 3D model itself with a CAD drawing and a PDF file containing the FEA report; you could also find a SolidWorks file depending on whether or not I had to modify the part in order to create the final assembly.

Every FEA considers the yield strength of ABS plastic:

$$\sigma_y = 20 \text{ MPa} \tag{1}$$

In the Base and last assembly FEA, we can see that the maximum calculated tension reaches about 60 MPa. However, that's not something to worry about because in both cases it occurs to be a singularity in the equation:

$$\sigma_{ij}(r,\theta) = \sum_{k=1}^{\infty} A_k r^{\lambda_k - 1} f_{ij}^{(k)}(\theta) \tag{2}$$

due to the edge not being smooth enough. That's something we don't have to take account of in our simulation because it's a merely simulative singularity. From the results, we can be sure the 3D structure will not fail under a 1 kg load. As stated before, that's more than the maximum load the SG90 can handle, so we are safe as we will never be able to lift that kind of load anyway.

4 Gripper Claw Mechanism

The end effector of the robotic arm features a custom-designed parallel mechanism.

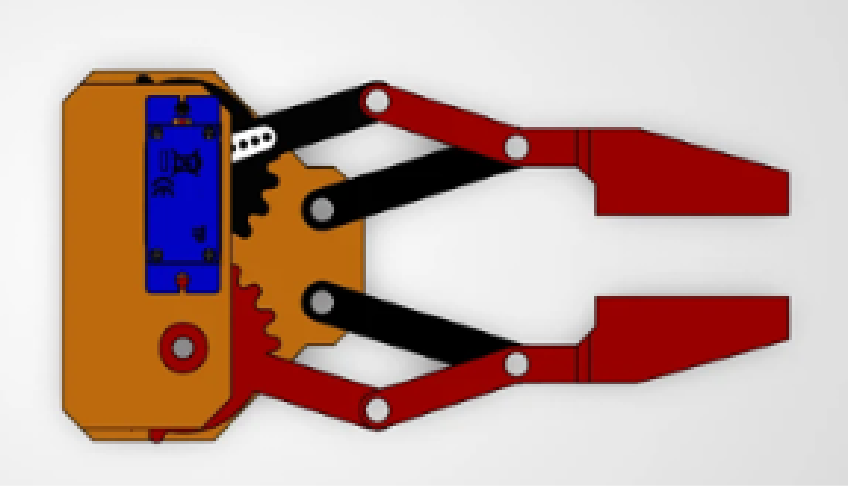


Figure 7: Top view of the gripper claw CAD model

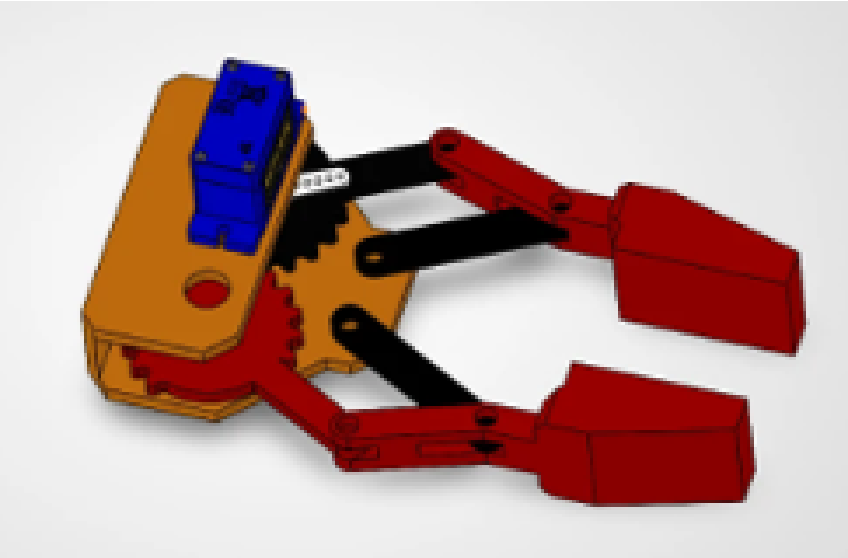


Figure 8: Isometric view of the gripper claw CAD model

The mechanism is driven by a servo attachment connected to an upper driving gear, which rotates with an angular velocity ω_1. This motion is transmitted to an identical lower driven gear, which consequently rotates in the opposite direction:

$$\omega_2 = -\omega_1 \tag{3}$$

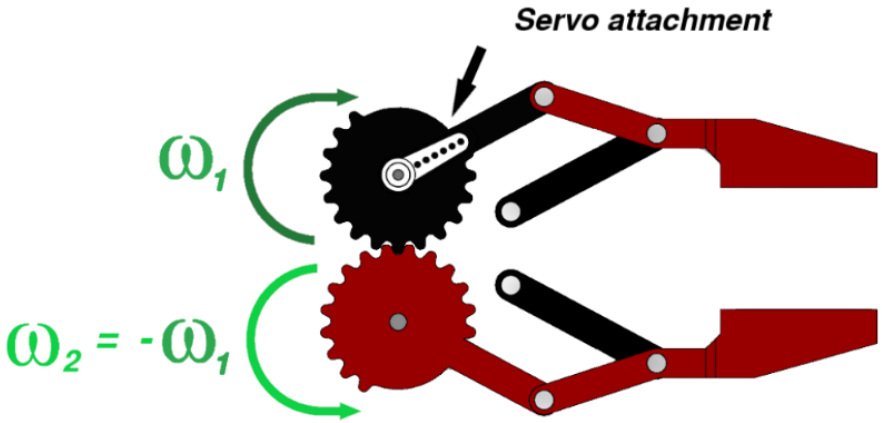


Figure 9: Kinematic diagram of the gripper claw mechanism

The leverage functioning relies on points forced to rotate with circular motion, described by the trajectory:

$$x^2 + y^2 = R \tag{4}$$

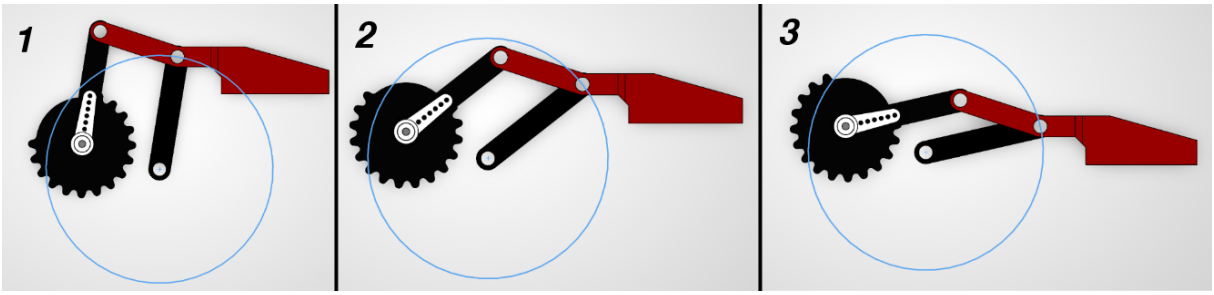


Figure 10: gripper claw mechanism circular motion

At the same time, the base pivots of the linkages are strictly locked, meaning their displacement over time is zero:

$$\frac{dx}{dt} = \frac{dy}{dt} = 0 \tag{5}$$

This specific type of leverage ensures that the two gripping faces remain always parallel to each other throughout the entire movement, from minimum to maximum excursion. This parallel geometry provides a stronger and tighter grip on the target objects. Furthermore, the design provides a large amount of space on the gripping surfaces, allowing for the attachment of high-grip materials such as rubber to further enhance the holding capacity.

5 v1 Prototype 3D Rendering

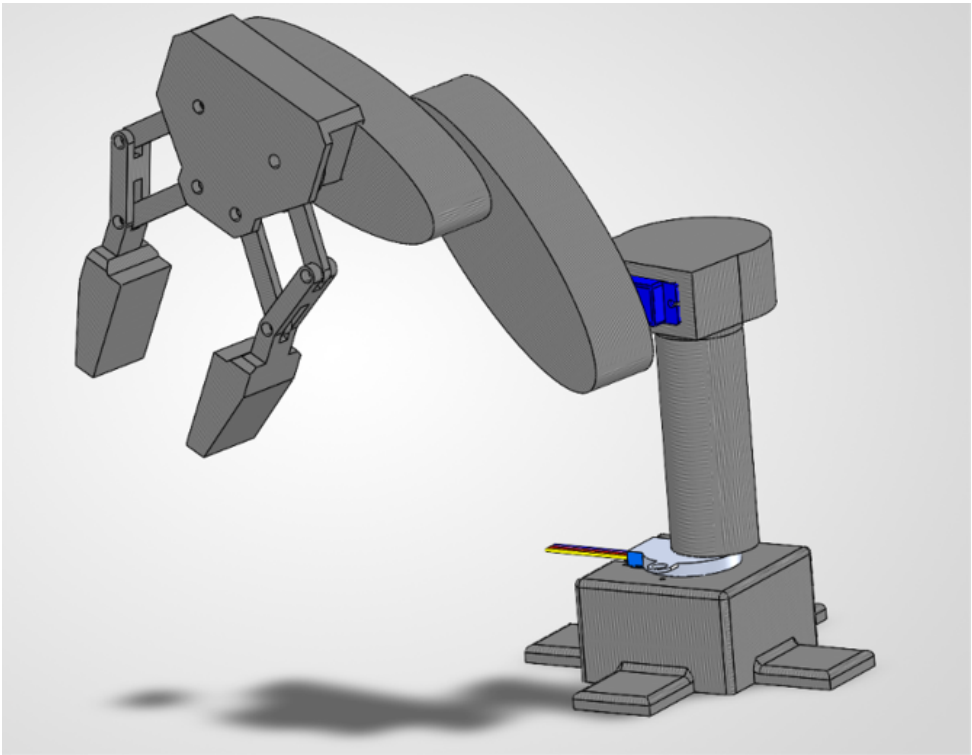


Figure 11: 3D rendering of the v1 prototype

Mechanical component

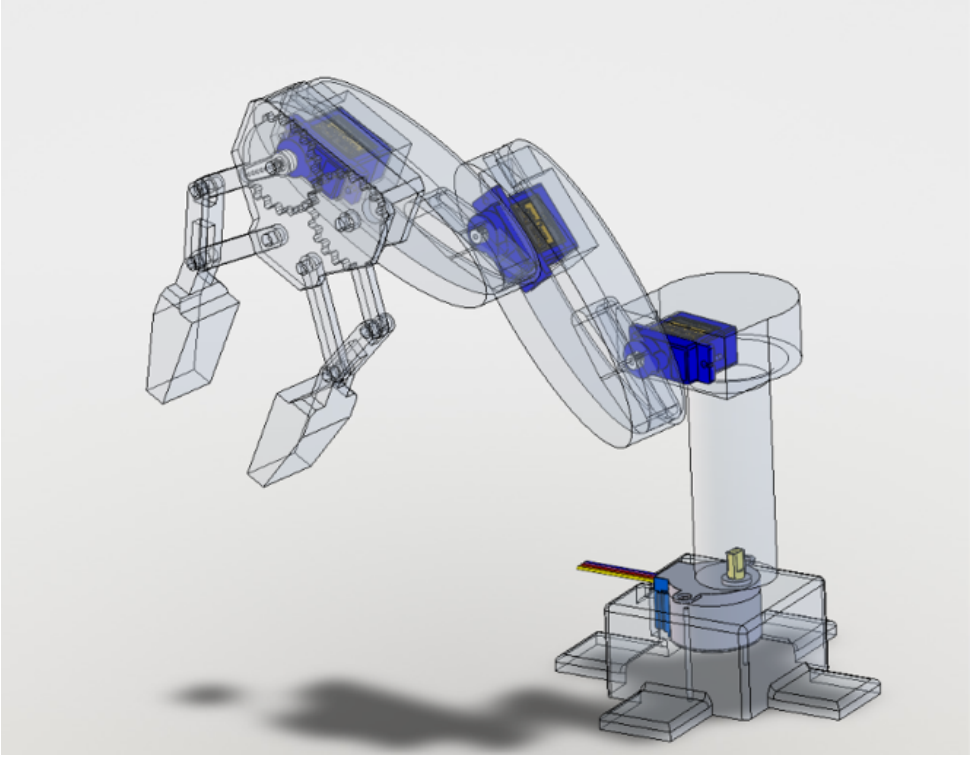


Figure 12: Mechanical components overview (transparent view)